

1) Find the derivative $\frac{dy}{dx}$.

a) $y = \sin(3x^2 + 2x + 7)$ $u = 3x^2 + 2x + 7$

$y = \sin(u)$

$du = 6x + 2$

$\frac{dy}{dx} = \cos(u) \cdot \frac{du}{dx}$

$\frac{dy}{dx} = \cos(3x^2 + 2x + 7)(6x + 2)$

Quotient Rule

c) $y = \frac{x+1}{x^2+1}$

$$\frac{dy}{dx} = \frac{(x^2+1)(1) - (x+1)(2x)}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{(x^2+1) - (2x^2+2x)}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{x^2+1-2x^2-2x}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{-x^2-2x+1}{(x^2+1)^2}$$

Product Rule Again

e) $y = (8x+1)^4(9x-3)^7$

$\frac{d}{dx}[fg] = f'g + fg'$

$f = (8x+1)^4$

$f' = 4(8x+1)^3(8)$

$f' = 32(8x+1)^3$

$g = (9x-3)^7$

$g' = 7(9x-3)^6$

$g' = 63(9x-3)^6$

$$\frac{dy}{dx} = [32(8x+1)^3][(9x-3)^7] + [(8x+1)^4][63(9x-3)^6]$$

$$\frac{dy}{dx} = 32(8x+1)^3(9x-3)^7 + 63(8x+1)^4(9x-3)^6$$

b) $y = \tan^{-1}\left(\frac{1}{x}\right)$ $\frac{d}{dx}[\tan^{-1}(u)] = \frac{u'}{1+u^2}$

$$\frac{dy}{dx} = \frac{-\frac{1}{x^2}}{1+\left(\frac{1}{x}\right)^2}$$

$$u = \frac{1}{x} = x^{-1}$$
$$\frac{du}{dx} = -x^{-2} = -\frac{1}{x^2}$$

$$\frac{dy}{dx} = \frac{-x^{-2}}{\left(1+x^{-2}\right)} = \frac{-\frac{1}{x^2+1}}{x^2+1}$$

Product Rule

d) $xy + x^2y^2 = 1$ $\frac{dy}{dx}[xy] + \frac{dy}{dx}[x^2y^2] = \frac{dy}{dx}[1]$

$$\left(x\frac{dy}{dx} + y\right) + \left(x^2 \underbrace{\frac{d}{dx}[y^2]}_{2y\frac{dy}{dx}} + y^2 \underbrace{\frac{d}{dx}[x^2]}_{2x}\right) = 0$$

$$x\frac{dy}{dx} + y + x^2(2y\frac{dy}{dx}) + y^2(2x) = 0$$

$$x\frac{dy}{dx} + y + 2x^2y\frac{dy}{dx} + 2xy^2 = 0$$

$$x\frac{dy}{dx} + 2x^2y\frac{dy}{dx} = -y - 2xy^2$$

$$\frac{dy}{dx}(x + 2x^2y) = -y - 2xy^2$$

$$\frac{dy}{dx} = \frac{-y - 2xy^2}{x + 2x^2y}$$

$$\frac{dy}{dx} = \frac{-y(1+2xy)}{x(1+2xy)}$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

f) $y = xe^{x^2}$

 $f'g + fg'$

$f = x$
 $f' = 1$

$g = e^{x^2}$
 $g' = e^{x^2}(2x)$

$$\frac{dy}{dx} = (1)(e^{x^2}) + (x)(e^{x^2} \cdot 2x)$$

$$\frac{dy}{dx} = e^{x^2} + 2x^2e^{x^2}$$

$$\frac{dy}{dx} = e^{x^2}(1 + 2x^2)$$

2) Find the antiderivatives.

a) $\int \sin^2 x \, dx$

$$\frac{\sin(2x)}{2} = \frac{2\cos(2x)}{2}$$

$$\frac{1}{2} \sin(2x) = \cos(2x)$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\begin{aligned} \int \sin^2 x \, dx &= \int \left(\frac{1 - \cos(2x)}{2} \right) dx \\ &= \frac{1}{2} \int 1 \, dx - \frac{1}{2} \int \cos(2x) \, dx \\ &= \frac{x}{2} - \frac{1}{2} \left(\frac{1}{2} \sin(2x) \right) + C \\ &= \frac{x}{2} - \frac{\sin(2x)}{4} + C \end{aligned}$$

b) $\int \frac{\sin x}{\cos^2 x} \, dx$

$$\frac{d}{dx} [\cos x] = -\sin x$$

$$u = \cos x$$

$$du = -\sin x \, dx$$

$$\hookrightarrow \sin x \, dx = -du$$

$$\int \frac{\sin x}{u^2} \, dx$$

$$\int \frac{\sin x \, dx}{u^2}$$

$$\int \frac{-du}{u^2}$$

$$= \int \frac{1}{u^2} \, du$$

$$= \int u^{-2+1} \, du$$

$$= \cancel{\frac{u^{-1}}{-1}} + C$$

$$= u^{-1} + C$$

$$= \cos^{-1} x + C$$

$$= \boxed{\sec x + C}$$

c) $\int \frac{x+1}{\sqrt{x^2+2x}} \, dx$

$$u = x^2 + 2x$$

$$du = 2x + 2 \, dx$$

$$du = 2(x+1) \, dx$$

$$(x+1) \, dx = \frac{du}{2} = \frac{1}{2} du$$

$$\int \frac{(x+1) \, dx}{\sqrt{u}}$$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du$$

$$\frac{1}{2} \int u^{-1/2+1} \, du$$

$$= \frac{1}{2} \cancel{2} u^{1/2} + C$$

$$= u^{1/2} + C = \boxed{\sqrt{x^2+2x} + C}$$

d) $\int \frac{x-1}{x^4} \, dx$

$$\int \frac{\cancel{1}x}{x^{4+3}} - \frac{1}{x^4}$$

$$\int \frac{1}{x^3} - \frac{1}{x^4}$$

$$\int \frac{x^{-3+1}}{-2} - \frac{x^{-4+1}}{-3}$$

$$= -\frac{1}{2} x^{-2} + \frac{1}{3} x^{-3} + C$$

$$= \boxed{-\frac{1}{2x^2} + \frac{1}{3x^3} + C}$$

e) $\int x \sin(x^2) \, dx$

$$u = x^2$$

$$du = 2x \, dx$$

$$\frac{1}{2} du = x \, dx$$

$$\int \sin(u) x \, dx$$

$$\int \sin(u) \frac{1}{2} du$$

$$\frac{1}{2} \int \sin(u) \, du$$

$$= \frac{1}{2} [-\cos(u)] + C$$

$$= \boxed{-\frac{1}{2} \cos(x^2) + C}$$

f) $\int x \sqrt{x+3} \, dx$

$$u = x+3 \Rightarrow x = u-3$$

$$du = 1 \, dx$$

$$\int (u-3) u^{1/2} \, du$$

$$\int \frac{u^{3/2+1}}{5/2} - 3 \frac{u^{1/2+1}}{3/2} \, du$$

$$= \frac{2}{5} u^{5/2} - 2 u^{3/2} + C$$

$$= \boxed{\frac{2}{5} (\sqrt{x+3})^5 - 2 (\sqrt{x+3})^3 + C}$$

3) Find the interval(s) where $f(x) = x^2 e^{-x}$ is concave down.

Concave down is:

$$f''(x) < 0$$

$$f''(x) = (-e^{-x})(2x - x^2) + e^{-x}(2 - 2x)$$

$$f''(x) = e^{-x} [-(2x - x^2) + (2 - 2x)]$$

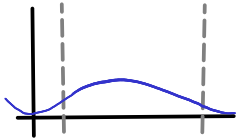
$$f''(x) = e^{-x} [-2x + x^2 + 2 - 2x]$$

$$f''(x) = e^{-x} (x^2 - 4x + 2) < 0 ?$$

$$f'(x) = 2x e^{-x} + x^2 (-e^{-x})$$

$$f'(x) = 2x e^{-x} - x^2 e^{-x}$$

$$f'(x) = e^{-x} (2x - x^2)$$



$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16 - 8}}{2}$$

$$x = \frac{4 \pm \sqrt{8}}{2} = \frac{4 \pm 2\sqrt{2}}{2} = 2 \pm \sqrt{2}$$

Because $x^2 - 4x + 2$ is an upward opening Parabola, it's negative between its roots:

$$(2 - \sqrt{2}, 2 + \sqrt{2}) \leftarrow \text{So this is where concave down}$$

4) Find the interval(s) where $f(x) = \frac{x+1}{\sqrt{x^2+1}}$ is decreasing.

Decreasing, so we need $f'(x) < 0$

$$f(x) = (x+1)(x^2+1)^{-1/2}$$

$$f'(x) = (1)(x^2+1)^{-1/2} + (x+1)\left(-\frac{1}{2}\right)(x^2+1)^{-3/2}(2x)$$

$$f'(x) = (x^2+1)^{-1/2} - x(x+1)(x^2+1)^{-3/2}$$

$$f'(x) = \frac{(x^2+1)^{-3/2} [(x^2+1) - x(x+1)]}{(x^2+1)^{3/2}}$$

$$\begin{aligned} &= \frac{1}{(x^2+1)^{3/2}} [x^2+1 - (x^2+x)] \\ &= \frac{1}{(x^2+1)^{3/2}} [x^2+1 - x^2 - x] \\ &= \frac{1-x}{(x^2+1)^{3/2}} \end{aligned}$$

$$f'(x) = \frac{(1-x)}{(x^2+1)^{3/2}} < 0$$

Only the numerator matters:

$$1 - x < 0$$

$$1 < x$$

$$x > 1$$

Any number that's bigger than 1:

$$(1, \infty)$$

5) Estimate $\int_1^2 e^{x^2} dx$ using the midpoints of eight rectangles.

$$\Delta x = \frac{b-a}{n}$$

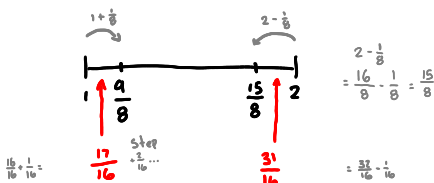
$$= \frac{2-1}{8} = \frac{1}{8} \quad (\text{half is } \frac{1}{16})$$

interval from 1 to 2

$\hookrightarrow 1 + \frac{1}{8} = \frac{9}{8}$ then $+\frac{1}{8} \dots$ etc

$$(1/8) \cdot \text{Sum}(\text{seq}(e^{(x/16)^2}, x, 17, 31, 2))$$

$$= 14.8527$$



6) A right circular cylinder must have a volume of 100 m^3 . Find the dimensions that should be used to make the cylinder using the least amount of material possible.

$$\begin{aligned}
 V &= 100 \text{ m}^3 \\
 V &= \pi r^2 h \\
 \pi r^2 h &= 100 \\
 h &= \frac{100}{\pi r^2}
 \end{aligned}$$

Surface Area $S = 2\pi r^2 + 2\pi r h$

$$\begin{aligned}
 S &= 2\pi r^2 + 2\pi r \left(\frac{100}{\pi r^2} \right) \\
 S &= 2\pi r^2 + \frac{200}{r}
 \end{aligned}$$

$$\begin{aligned}
 S'(r) &= 4\pi r - \frac{200}{r^2} = 0 \\
 4\pi r &= \frac{200}{r^2} \\
 4\pi r^3 &= 200 \\
 2\pi r^3 &= 100 \\
 r^3 &= \frac{50}{\pi} \\
 r &= \sqrt[3]{\frac{50}{\pi}}
 \end{aligned}$$

$$\begin{aligned}
 h &= \frac{2\pi r^3}{\pi r^2} \\
 h &= \frac{2r^3}{r^2} = 2r \\
 h &= 2r
 \end{aligned}$$

$$\begin{aligned}
 h &= 2 \left(\sqrt[3]{\frac{50}{\pi}} \right) \text{ m} \\
 r &= \sqrt[3]{\frac{50}{\pi}} \text{ m}
 \end{aligned}$$

7) A rectangular swimming pool has a base that is 10m by 20m. Water is being pumped into the pool at a rate of 30 m^3 . How fast is the water level rising after 12 minutes.

10 m by 20m

$$30 \text{ m}^3/\text{min} \rightarrow \frac{dV}{dt} = 30 \text{ m}^3/\text{min}$$

Pool Volume: $V = lwh$, $l = 20$, $w = 10$

$$V = (20)(10)h$$

$$V = 200h$$

$$\frac{dV}{dt} = 200 \frac{dh}{dt}$$

$$200 \frac{dh}{dt} = 30 \text{ m}^3/\text{min}$$

$$\frac{dh}{dt} = \frac{3}{20} = 0.15 \text{ m/min}$$

8) Use Newton's method to estimate the smallest positive root of $f(x) = -x^3 + 2x^2 - 5x + 1$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x) = -3x^2 + 4x - 5$$

We want the smallest Positive root possible
Since $f(0) = 1$ and $f(1) = -1 + 2 - 5 + 1 = -3$

There is a positive root between 0 and 1
Start with $x_0 = 0$

$$x_1 = 0 - \frac{f(0)}{f'(0)}$$

$$x_1 = 0 - \frac{1}{-5}$$

$$x_1 = 0.2$$

$$x_2 = 0.2 - \frac{f(0.2)}{f'(0.2)}$$

etc... until the last two results are similar or the same.

0 Enter

$$\text{Ans} - ((- \text{Ans}^3 + \text{Ans}^2 - 5 \text{Ans} + 1) / (-3 \text{Ans}^2 + 4 \text{Ans} - 5))$$

$$\text{Enter} = 0.2 \quad (x_1)$$

$$\text{Enter} = 0.2166667 \quad (x_2)$$

$$\text{Enter} = 0.2167566 \quad (x_3)$$

$$x = 0.217$$

9) Evaluate the limits. Be sure to show all work.

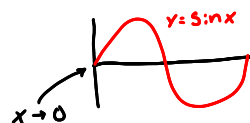
$$\text{a) } \lim_{x \rightarrow \infty} \frac{5x^2 - 8x + 1}{2x^2 + 6x - 1}$$

Since the highest power is x^2 ,

$$\lim_{x \rightarrow \infty} \frac{\frac{5x^2}{x^2} - \frac{8x}{x^2} + \frac{1}{x^2}}{\frac{2x^2}{x^2} + \frac{6x}{x^2} - \frac{1}{x^2}}$$

$$\lim_{x \rightarrow \infty} \frac{5 - \frac{8}{x} + \frac{1}{x^2}}{2 + \frac{6}{x} - \frac{1}{x^2}} = \frac{5}{2}$$

$$\text{b) } \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x}}$$



$\sin(0) = 0$, So as we approach 0^- and 0^+ the numerator gets smaller and smaller.

So near zero, $\sin x \approx x$

$$\rightarrow \frac{\sin x}{\sqrt{x}} \approx \frac{x}{\sqrt{x}}$$

$$= \frac{x}{x^{1/2}}$$

$$= x(x^{-1/2})$$

$$= x^{1-1/2}$$

$$= x^{1/2}$$

$$= \sqrt{x} \quad (x \text{ can only be positive})$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0$$